COMP 421 - Database Systems

Project 1: Database Design and Data Modelling

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Assumptions

Restrictions

Relational Translation

`HealthcareInstitution(hcId, phoneNum, name, email, address, website)`

`CommunityClinic(hcId)`

`hcId` foreign key referencing `HealthcareInstitution`

`BirthingCenter(hcId)`

`hcId` foreign key referencing `HealthcareInstitution`

`Midwife(pracId, name, email, phoneNum, hcId)`

`hcId` foreign key referencing `HealthcareInstitution`

note: `hcId` has to be `NOT NULL` to enforce Midwife's participation constraint in the `employs` relationship set.

`Father(fatherId, phoneNum, name, email, address, QChcardid, dateOfBirth, profession, bloodType)`

`Mother(QChcardid, phoneNum, name, email, adddress, dateOfBirth, profession, bloodType)`

`Couple(coupleId, invited, QChcardid, fatherId)`

`QChcardid` foreign key referencing `Mother`
`fatherId` foreign key referencing `Father`

note: `QChcardid` has to be `NOT NULL` to enforce Couple's participation constraint in the `constitutesF` relationship set.

`InformationSession(sessId, language, date, time, pracId)`

pracId foreign key referencing Midwife

note: pracId has to be NOT NULL to enforce InformationSession's participation constraint in the hosts relationship set.

Invitation(sessId, coupleId, attended, registered)

Pregnancy(coupleId, pregNum, estDueDate, expBirthTimeframe, lastMenstPeriod, isHomeBirth, primaryMidwife, backupMidwife)

coupleId foreign key referencing Couple
primaryMidwife foreign key referencing Midwife(pracId)
backupMidwife foreign key referencing Midwife(pracId)

note: coupleId is part of the primary key and thus has to be NOT NULL.

Appointment(apptId, date, time, pregNum, coupleId, pracId)

(pregNum, coupleId) foreign key referencing Pregnancy
pracId foreign key referencing Midwife

note: pregNum and coupleId has to be NOT NULL to enforce Appointment's participation constraint in the belongs relationship set.

note: pracId has to be NOT NULL to enforce Appointment's participation constraint in the sets up relationship set.

Note(noteId, date, time, description, apptId)

apptId foreign key referencing Appointment

note: apptId has to be NOT NULL to enforce Note's participation constraint in the belongs relationship set.

Baby(babyId, name, gender, timeOfBirth, dateOfBirth, bloodType, pregNum, coupleId)

(pregNum, coupleId) foreign key referencing Pregnancy

note: pregNum and coupleId has to be NOT NULL to enforce Baby's participation constraint in the born from relationship set.

Technician(techId, name, phoneNum)

Test(testId, type, dateSampleTaken, result, dateConducted, lab, techId, pracId, pregNum, coupleId, babyId, QChcardid)

techId foreign key referencing Technician
pracId foreign key referencing Midwife
(pregNum, coupleId) foreign key referencing Pregnancy
babyId foreign key referencing Baby
QChcardid foreign key referencing Mother

note: pregNum, coupleId has to be NOT NULL to enforce Test's participation constraint in the belongs relationship set.

note: `techId` has to be `NOT NULL` to enforce Test's participation constraint in the `conducts` relationship set.

note: `pracId` has to be `NOT NULL` to enforce Test's participation constraint in the `prescribes` relationship set.

BirthLocation(hcId, pregNum, coupleId) `hcId` foreign key referencing BirthingCenter

`(pregNum, coupleId)` foreign key referencing `Pregnancy`